Q1.

Two different companies manufacture a material. A researcher collected samples from both companies, each of size n=25

- Company 1: $\bar{X}_1=20$ milligrams/1000 cycles, $S_1=2$ milligrams/1000 cycles—Company 2: $\bar{X}_2=15$ milligrams/1000 cycles, $S_2=8$ milligrams/1000 cycles
- Given that the population variances are **not assumed equal** and significance level is $\alpha=0.05$ Test whether the data support the claim that the companies produce material with different mean wear.

Degrees of Freedom v (Welch's approximation):

$$v = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2-1}}$$

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Q1

Data:

Sample size : $n = 25$

Company 1:

$\bar{x}_1 = 20$ milligrams/1000

$S_2 = 2$ milligrams/1000

Company 2:

$\bar{x}_2 = 15$ milligrams/1000

$S_2 = 8$ milligrams/1000

$\alpha = 0.05$.

$\sigma_1^2 \neq \sigma_2^2$ (given)

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To find:

Do data support the claim that companies produce material with different mean wear.

Solution:

⇒ **Hypothesis:**

$$\begin{aligned} H_0 &: \mu_1 = \mu_2, \quad \left\{ \mu_1 - \mu_2 = \overset{d_0}{0} \right. \\ H_1 &: \mu_1 \neq \mu_2, \quad \left\{ \mu_1 - \mu_2 \neq 0 \right. \end{aligned}$$

=) Test statistics:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$= \frac{(20-15)-0}{\sqrt{\frac{(2)^2}{25} + \frac{(8)^2}{25}}} = \frac{5}{\sqrt{\frac{4}{25} + \frac{64}{25}}}$$

$$\boxed{t = 3.0316}$$

=) Critical Region

$$\alpha = 0.05 \quad \frac{\alpha}{2} = 0.025$$

$$v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

$$= \frac{(4/25 + 64/25)^2}{\frac{(4/25)^2}{24} + \frac{(64/25)^2}{24}}$$ 26.9

$$= \frac{7.3}{(1.066 \times 10^{-3}) + (0.2730)} \quad \boxed{v = 26.6} \simeq 27$$

value with $\alpha = 0.025$ and $v = 27$ is 2.052

$$\left.\begin{array}{l} t \geq t_{\alpha/2,(v)} \\ t \leq t_{\alpha/2,(v)} \end{array}\right\} \text{conditions.}$$

$$\left.\begin{array}{c} 3.0316 \geq 2.052 \text{ (True)} \\ \text{OR} \\ 3.0316 \leq -2.052 \end{array}\right\} \text{Checking}$$

H_0 Rejected.

Conclusion:

There is no evidence to say that two companies produce material with same mean, as Test stats is inside ~~outside~~ critical region.

Q2.

A survey of 500 employees was conducted to examine the relationship between their **job classification** (Salaried or Hourly) and their **preference for one of three pension plans** The results are shown below:

Job Classification	Pension Plan 1	Pension Plan 2	Pension Plan 3
Salaried	160	140	40
Hourly	40	60	60

The significance level is α=0.05**.** Test whether **job classification** and **preference for pension plan** are **independent**.

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(Q2)

Data:

$N = 500$ (Sample Size)

$\alpha = 0.05$

To find:

evaluate independence between preference of plans and job classification

Sol:

[redacted] ei's (expected freq)

Pension Plan

Job classification	1	2	3	Total
Salaried	160	140	40	340
Hourly	40	60	60	160
	200	200	100	500

Table for ei's (expected frequency Table)

$$e_i = \frac{(\text{Row Total})(\text{Col Total})}{\text{Grand Total}}$$

Pension Plan

Job classification	1	2	3
Salaried	136	136	68
Hourly	64	64	32

Hypothesis:

- H_0: Preference of plans and job classification are independent
- H_1: Preference of plans and job classification are dependent.

Test Stats:

$$\chi^2 = \sum_{i=0}^{i=n} \left[\frac{(O_i - e_i)^2}{e_i} \right]$$

$$= \frac{(160-136)^2}{136} + \frac{(140-136)^2}{136} + \frac{(40-68)^2}{68}$$

$$+ \frac{(40-64)^2}{64} + \frac{(60-64)^2}{64} + \frac{(60-32)^2}{32}$$

$$\boxed{= 49.632}$$

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Critical Region:

$\chi^2 > \chi^2_{\alpha}(v)$] condition

$v = \underset{\text{No. of row}}{(NR-1)} \underset{\text{No. of col}}{(Nc-1)} = (1)(2)$

$v = 2$

value at $\alpha = 0.05$ $v = 2$ is $\chi^2_{0.05(2)} = 5.991$

Check: $49.632 > 5.991$ (True) H_0 Reject

Conclusion:

There is no evidence to say that preference of plans and job classification are independent as Test stats is in critical region.

Q3.
Sample size: n=9.Data for x and y are given.

(a) Find the **linear regression line** of y on x.

(b) Estimate y^when x=85.

$$b_1 = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

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(Q3)

Data:

$n = 9$

x, y given

To find:

linear regression line.

$\hat{y}$ at $x = 85$.

Sol:

Linear regression line

$$\hat{y} = \underbrace{b_0}_{\text{intercept}} + \underbrace{b_1}_{\text{slope}} x$$

$$b_1 = \frac{SSxy}{SSxx}$$

$$SSxx = \sum x^2 - \frac{(\sum x)^2}{n}$$

$$= 57557 - \frac{(707)^2}{9}$$

$$= 2018.22$$

$$SSxy = \sum xy - \frac{\sum x \sum y}{n}$$

$$= 53258 - \frac{(707)(658)}{9}$$

$$= 1568.444$$

$$b_1 = \frac{1568.444}{2018.22} \qquad \boxed{b_1 = 0.7771}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$
$$= (73.111) - (0.7771)(78.55)$$

$$\boxed{b_0 = 12.06}$$

$\bar{y} = \frac{658}{9} = \frac{\Sigma y}{n}$
$= 73.111$

$\bar{x} = \frac{707}{9} = \frac{\Sigma x}{n}$
$= 78.55$

Linear Regression line

$$\hat{y} = b_0 + b_1 x$$

$$\Rightarrow \hat{y} = 12.06 + (0.7771)x \quad -(1)$$

Part (b)

85 on mid
So $x = 85$
put in eq (1)

$$\hat{y} - 12.06 + (0.7771)(85)$$

$$\boxed{\hat{y} = 78.1135}$$

final exam grade of student with 85 on mid.